



MANIPAL
UNIVERSITY JAIPUR
(University under Section 2(f) of the UGC Act)



INTERNATIONAL INNOVATION CHALLENGE 3.0



Theme: Cybersecurity & Digital



Sovereignty

SATARK · India's First Anti-Digital-Arrest Shield



Recognize → Interrupt → Verify → Package → Report

Team: 511

₹4,057.7 cr lost · 2,97,727 complaints · zero consumer defenses (2022 to May 2026)

INDIA'S LARGEST QUANTIFIED FRAUD • ₹4,057.7 CRORE

AND THE VICTIM'S PHONE IS UNGUARDED

₹4,057.7 cr

lost to digital-arrest scams (2022 • May 2026)

2,97,727

complaints filed • I4C / NCRP

3×

complaint growth 2022 → 2024

HOW THE SCAM WORKS

- 01** Fake 'CBI / police' call • arrest threat, stay on the line
- 02** Victim isolated • forbidden from telling anyone
- 03** Life savings moved via UPI within minutes
- 04** The truth surfaces hours later • or never

Fraud completes at ONE moment • the UPI payment. Nobody guards it.

SATARK • THE PRODUCT LOOP

Recognize → Interrupt → Verify → Package → Report

RECOGNIZE

INTERRUPT

VERIFY

PACKAGE

REPORT

ON THE CALL • BEFORE THE PIN

- 1 AI flags scam talk + pressure • in Hindi
- 2 PAUSE screen fires • call held, elder warned
- 3 Family confirms, challenges, or kills the payment

AFTER THE CALL • EVIDENCE

- 1 One-tap report to 1930 / Chakshu
- 2 Tamper-proof evidence packet (sha256 chain)
- 3 Timestamps, transcript spans, risk reasons
- ★ Premium tier: cloud-first-answer • OTP-bot proof

India's first call-state + UPI-tripwire + evidence fusion

THE MARKET • AND THE GRAVEYARD

₹22,846 cr lost in 2024 alone • 2.27M complaints • every incumbent misses the moment

2,268,346

cyber-fraud complaints 2024 • 5× in 3 years

₹22,845.7 cr

reported losses 2024 • 10× in 3 years

137.9 M

elders 60+ • 85.5% of households own a smartphone

47%

Indians hit by AI voice-clone scams • 83% lost money (McAfee)

THE ELDER-FINTECH GRAVEYARD

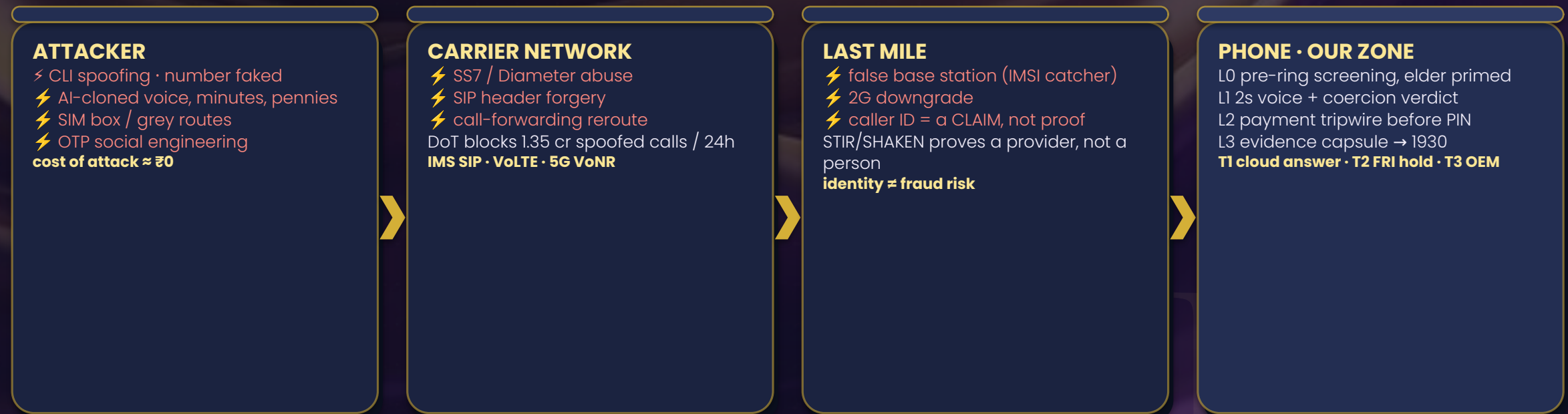
- ✗ Charlie • \$23M raised, DEAD (Jan 2026)
- ✗ Kalgera • pivoted consumer → B2B
- ✗ True Link • SURVIVED 250K+ families: control in the payment path

WHO'S IN THE LANE

- ☒ **Truecaller • 500M MAU**
caller ID only • no audio, no coercion semantics
- ☒ **Pixel / Galaxy Scam Detection**
English-first, device-locked, no evidence
- ☒ **Hiya / Pindrop**
enterprise-only, English, no India consumer product
- ☒ **SeniorShield.ai**
1K+ downloads, user-invoked, no fusion
- ☒ **PhonePe Protect**
no call context • acts AFTER the transaction
- ★ **SATARK • 4.18% EER vs 36 to 64% generic**
no public Hindi anti-spoof benchmark (900+ citations checked)
Attacker tooling: ElevenLabs 1-2 min • GPT-SoVITS 5s • RVC real-time

WHERE THE CALL GOES • AND WHERE IT GETS HACKED

Every hop is a trust boundary • nobody owns the phone moment but us



WHO ELSE COULD STOP IT HERE? Truecaller • caller ID only, hears nothing • Carriers • block numbers, not voices • Banks • act AFTER money moves
 NOBODY owns the seconds between 'scam detected' and 'payment made' • that gap is Satark



Our loop runs on every hop we can reach: pre-ring (L0) → first seconds (L1) → payment (L2) → hang-up (L3) • cloud tier (T1)

TECHNICAL APPROACH · BUILT & FEASIBLE

Measured, not promised · and honest about the platform wall

THE STACK

Python

PyTorch

AASIST

faster-whisper (Hindi)

Coercion lexicon

sha256 evidence

THE FLOW

audio

anti-spoof

coercion

verdict

PAUSE

evidence

1930 report

<500ms

warm verdict · measured

0.9919

held-out acc · 0/100 FP

66.3ms

per call · 3-crop vote

THE WALL · and how we beat it: consumer apps can't record live calls (VOICE_CALL is privileged).

We own the moments Android allows: pre-ring · payment screen · notification · post-call file · L0 to L4 + CPaaS tiers

IMPACT · SECONDS TO INTERVENTION

Measured on outcomes, not model scores

SECONDS

to intervention · vs minutes-to-loss. Every second is a payment that didn't move. The ₹4,057.7cr runway is the market.

0.37%

of ₹56,087cr reported is ever refunded · prevention is the only answer

RBI 2027 liability shift: banks become buyers of this evidence

SOCIAL

elders stay independent · families stay connected

ECONOMIC

feeds ₹7,130cr-saved reporting rails
(1930 / FRI)

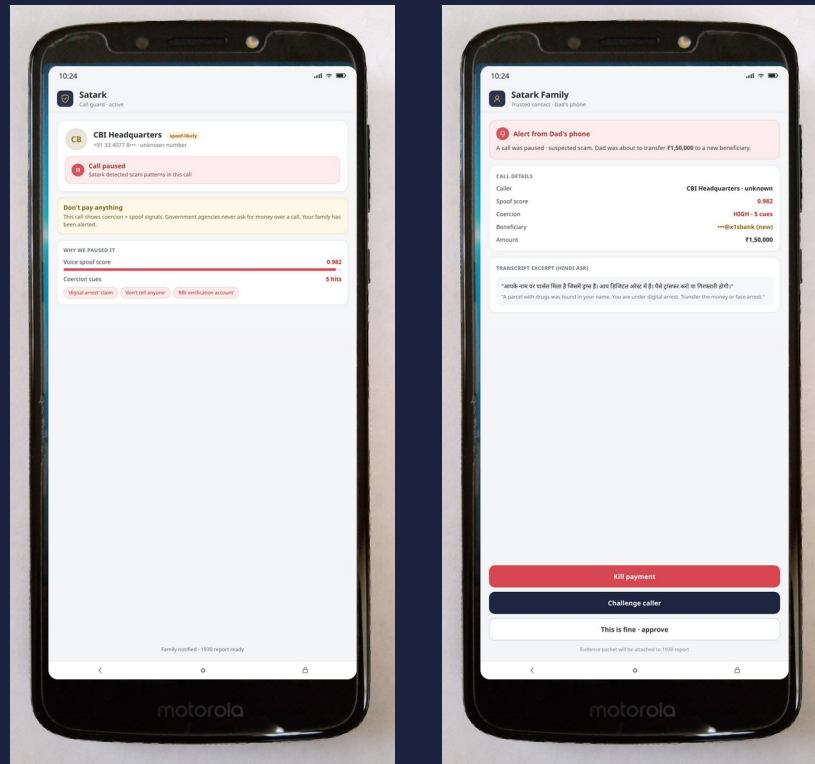
NATIONAL

digital sovereignty · DPDP-compliant by design

WORKING PROTOTYPE · ENGINE REAL, UI IN BUILD

What exists is real and measured · the screens show the spec, on a real device

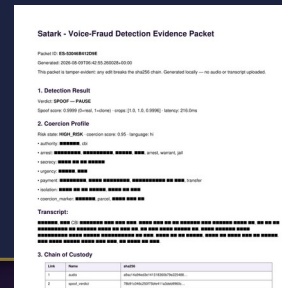
CONCEPT UI



ENGINE · BUILT & VERIFIED

0.9919 acc · 0/100 FP · <500ms

AASIST-hindi · 3-crop majority vote · held-out eval, not field



EVIDENCE PACKET

sha256 chain-of-custody → 1930-ready PDF
12 packets generated by the fusion battery

github.com/harshgounder/eldersshield

engine · coercion · evidence · battery tests · public repo (click to open)

Real device photo · spec screens composited. Final design under build.

RESEARCH AND REFERENCES

1. [RBI/2024-25/105 · voice/SMS fraud safeguards \(Jan 2025\) · rbi.org.in](#)
2. [RBI Auth Directions 2025 · 2FA mandatory Apr 2026 · rbi.org.in](#)
3. [CERT-In CIAD-2024-0060 · Deepfakes: threats & countermeasures · cert-in.org.in](#)
4. [DoT FRI: 1000+ banks · ₹660 crore prevented \(Dec 2025\) · pib.gov.in](#)
5. [SC order Aug 4 2026 · digital-arrest complaints 1,23,672 → 16,377 \(I4C data\) · thehindubusinessline.com](#)
6. [ASVspoof 2021 LA + NASK real-fraud benchmark \(WACV 2026\) · zenodo.org](#)

SATARK · THE SHIELD EVERY INDIAN FAMILY DESERVES

Recognize · Verify · Package · Report · Remediate